

RealityReceipt

Build spec and submission plan for Claude Code. HackGT 13, Georgia Tech, September 25 to 27, 2026.

One line: Every store sorts by sticker price. RealityReceipt sorts by what a purchase actually costs you: the price today, how you pay for it, what it costs to run and keep up, and how long it lasts, for new and used options side by side.

How to use this document

- This is the single source of truth for the build. Where it conflicts with older notes, this wins.
- Items marked **STATED** come from the official prize list or the team's plan. Items marked **VERIFY** are the team's working assumptions that must be checked against the primary source before they are coded as fact or said on stage.
- Build rule above all others: **only the first step uses AI**. Everything after extraction is deterministic and unit-tested.
- Writing style for any user-facing copy: no em dashes.

1. Event rules and what we are entering

- **STATED:** Teams up to 4; each winning member gets their own prize.
- **STATED:** Submit to exactly ONE themed track, plus any sponsor challenges we genuinely qualify for. Organizers verify sponsor eligibility.
- Grand prizes: 1st Canon G7X Mark III, 2nd Nintendo Switch 2, 3rd Apple iPad 11-inch.

Main track: A Marina's Mission (social good), presented by Aramco Americas

Official description	How RealityReceipt meets it
"Design technology that makes a real difference in our world, whether it's health, climate, energy, education, or a problem that hits close to home." Prizes: Bose QC / Ninja CREAMi.	Real difference: targets the poverty premium, where the option that looks cheapest today (old used goods, rent-to-own, high-interest credit) costs the most over time. Lower-income households are most exposed to it. Energy and climate: every path shows running energy cost and a carbon line, so the cheapest-to-own choice and the lower-emission choice are visible together. Open the pitch on energy, since Aramco Americas presents the track. Close to home: no sign-up, no income or personal questions. Evidence opens with a weighted analysis of EIA RECS 2020 household data.

Sponsor challenges we are entering

Challenge (as stated)	How we meet it	Must-do for eligibility
Visa: commerce experience powered by generative AI; make shopping intuitive, personalized, frictionless; secure and trusted payments; transform one or more stages (discovery, decision-making, personalization, checkout, payments, loyalty and rewards, post-purchase). \$5,000, first place only.	Sort by true cost. New and used options ranked by total cost and cost per year of use, not sticker price. Grok parses natural requests ("about \$300, small space, need it this week") into filters. Payment-path comparison (cash, card, BNPL, credit union PAL, rent-to-own) covers the payments stage. Covers discovery, personalization, decision-making, payments, post-purchase.	Lead the write-up with the generative AI shopping flow and used vs new ranking. Checkout links out to the retailer; never fake a payment. Known weak spot: our AI layer is intentionally small, so make the AI shopping step visible in the demo.
SpaceXAI: build with Cursor using Grok products (Grok models, Grok Imagine, Grok Voice) on real-world societal issues such as public health, sustainability, education. Custom Cursor keyboards; credits for all.	Grok is the whole AI layer: vision reads labels, price tags, leases and listing screenshots; Grok parses shopping requests; Grok Voice reads the receipt aloud in English and Spanish. Societal issue: the poverty premium and household energy costs.	Build in Cursor. Grok vision must be the primary path in the demo. OCR fallback (OpenCV + PaddleOCR) is a safety net only; if it carries the demo, eligibility is weak.
Notability: use Notability Pro during the event, tag the Devpost "Notability", add a note on how it was used with at least 2 screenshots. 1 year of Notability Pro each.	Use Notability for architecture sketches, the pitch storyboard and the demo run sheet.	Start using it now. Capture at least 2 screenshots. Tag Devpost and add the note.
Create-X: optional checkbox at submission for interest in their startup program.	Optional.	Tick the box at submission if the team is interested.

Conditional and skipped

- **TigerData: UNCONFIRMED.** Not in the official list. Enter only if a real prize is confirmed. If it is: store scans as time series and load EIA monthly Georgia residential electricity prices to project rates forward instead of holding them flat. Swap SQLite for TigerData Postgres behind the same repository interface.
- **Skipped:** Meta, Impiricus, NSA C1 HEARSAY, NSA C3 Codebreaker. Anyone free may try NSA C2 Packet Pursuit on the side. It is unrelated to this build.

2. The product

Problem

When people need to buy something, the options that look cheap often are not. An old used item can cost far more to run and keep up; rent-to-own commonly totals several times retail; repairing what you have may be cheapest today and most expensive a few years out. Shopping sites sort by sticker price, which hides all of this.

Scope: any purchase, not specific products

- The engine is category-agnostic. A category is a **data profile**, not code. Adding a category means adding data.
- Total cost of ownership covers: price today by payment path, running costs (energy, fuel, consumables), scheduled upkeep, repairs, and end of life (resale or disposal).
- Example of the breadth, not a focus: for a vehicle, running costs are fuel, upkeep is routine servicing on a schedule, repairs are ranges. Same engine, different profile.

- Categories with a federal energy label are the best-sourced case (certified per-model energy data), so they are what the demo uses. The pitch never centers on a specific product.

What the user does

- Scan or upload a rating label, price tag, lease, or listing screenshot. Or type a model number by hand.
- No sign-up. No income or personal questions.
- Optional "I can spend up to \$___ today" slider.

What the user gets: the receipt

Paths, each with three headline numbers, sorted by true cost:

Path	Notes
1. Repair the one you have	Repair cost as a labeled range.
2. Used, as-is	Price from the user's listing. Condition unknown, so ranges stay wide.
3. Used, refurbished	Warranty term shown on the receipt; narrows the range.
4. New	Sub-paths: cash, credit card, buy now pay later, credit union Payday Alternative Loan (PAL).
5. Rent-to-own	Keep paying to the end, or the cheapest early buyout point.

Headline number	Definition
Pay today	Cash out of pocket now for that path.
Total over 3 years	All costs in the 36-month window, including a replacement purchase if the item is expected to wear out inside the window.
Cost per year of use	Total cost divided by expected years of life (a range). This is the fair used vs new number: a cheap used item with little life left can cost more per year than a new one. If no published lifespan exists, show "not estimated" and fall back to the 3-year total.

Used vs new comparison

- New prices: cached retailer data, with source and retrieval date shown.
- Used prices: only from the listing the user scans or uploads. No scraping. No invented "market average".
- Remaining life: age from the listing or decoded serial, against a published lifespan range for the category. Shown as a range.

Trust layer: every number is tappable

Each cost line opens to its source and formula, and carries exactly one source label:

Label	Meaning
Rated	Certified for that exact model (for example, an energy certification dataset).
Published	From a manufacturer schedule or a government dataset for the category.
You entered	Typed by the user, such as a repair quote they received.
Not estimated	No defensible source. Left visibly blank. Never filled by AI or guessed.

Estimated figures (for example, energy for an old unit decoded from its age) are shown as ranges, never as a single point. Carbon line appears only where the item uses energy or fuel; hidden otherwise.

3. Architecture

Pipeline

```
photo -> vision extract (strict JSON) -> validate + manual correction -> serial decode ->
category profile lookup -> cost engine (36 months) -> receipt
```

- **Only the first step uses AI.** Everything after is deterministic and unit-tested.
- **"Not a wrapper" test (must pass):** typing brand, model and serial by hand produces the identical receipt as scanning. The AI is the keyboard.
- Vision output must validate against a strict Pydantic schema. On failure, show the manual correction form pre-filled with whatever parsed.
- Serial date codes are set by manufacturer, not product type, so decoders are keyed by brand and reused across categories. Some brands' year codes repeat on a cycle: resolve from model era, or set year_confidence low and flag it.

Category profile (data, not code)

- Running-cost model: rated energy source (energy certification dataset first, then DOE Compliance Certification Database), standard usage assumption from the federal label, optional usage adjust slider (never required).
- Estimated energy for old units: decoded manufacture year, then the DOE efficiency standard in force that year, shown as a range.
- Upkeep schedule (published only), repair cost ranges, lifespan range, end-of-life notes.
- Any field without a source is stored as not_estimated.

Cost engine rules

- Electricity: Georgia Power residential **marginal** rate from the tariff sheet, not the blended average. Include per-kWh riders; document tier and season assumption.
- Carbon: EPA eGRID emission rate for Georgia's subregion. Record which rate type is used.
- Credit card: Federal Reserve G.19 average rate.
- PAL (NCUA PALS II, STATED): up to \$2,000, 1 to 12 months, interest capped at 28%, application fee capped at \$20. Show as an option with caps; never imply the user can get one.
- Rent-to-own: full payment stream, cheapest early buyout week, and "effective annual cost". **Never label it APR.** Prefer the scanned lease's own numbers (total of payments vs cash price) over any generic multiple.
- Repairs as labeled ranges. **No invented failure probabilities.**
- 36-month monthly cash flow per path; replacement purchase inserted if expected life ends inside the window.

API

```
POST /scan {kind: label|price_tag|lease|listing, image} -> extracted JSON
POST /item manual or corrected item -> Item
POST /quote {item_id | items[], budget_today?, usage_adjust?} -> Path[]
POST /shop/parse natural-language request -> filters (Grok)
POST /shop/rank filters + candidate offers -> offers ranked by true cost
GET /categories available category profiles
GET /sources all sources with id, publisher, url, retrieved date
```

Data model (category-agnostic)

Item: id, category, brand, model, serial, mfg_year, year_confidence, condition (new|used_as_is|refurbished), warranty_months, attributes{}

Offer: item_id, price, seller_type, source (retailer_cache|user_listing|price_tag), retrieved_at

Lease: weekly_payment, term_weeks, cash_price, fees, early_purchase_rule, missed_payment_rule

CostLine: kind (purchase|financing|running|upkeep|repair|replacement|end_of_life), label, amount_low, amount_high, period, source_type (rated|published|user_entered|not_estimated), source_id, formula

Path: name, payment_method, pay_today, total_3yr_low, total_3yr_high, cost_per_year_low, cost_per_year_high, expected_life_low, expected_life_high, monthly[36], carbon_kg (nullable), lines[CostLine], flags[]

CategoryProfile: category, energy_dataset_refs, usage_assumption, upkeep_schedule[], repair_ranges[], lifespan_range, carbon_applicable

Source: id, title, publisher, url, retrieved_date, notes

Stack

- Frontend: React + Vite PWA, getUserMedia camera, Tailwind, Framer Motion receipt animation.
- Backend: FastAPI + Pydantic. SQLite preloaded with all datasets (TigerData Postgres only if that prize is confirmed). pytest on the engine.
- AI: Grok vision primary; OpenCV cleanup + PaddleOCR fallback. Grok Voice for read-aloud (English, Spanish). Built in Cursor.
- Phone cameras need HTTPS: deploy, or use a cloudflared tunnel. Test on a real phone early.
- **Nothing fetched live during the demo.** All datasets and retailer results cached.

4. Data sources

Source	Use	Status
ENERGY STAR certified datasets (data.energystar.gov)	Rated per-model energy	STATED
DOE Compliance Certification Database	Rated energy fallback	STATED
DOE efficiency standards by year	Estimated energy ceiling for old units	STATED; see verify list
Georgia Power residential tariff sheet	Marginal electricity rate	To pull
EPA eGRID (Georgia subregion)	Carbon per kWh	STATED
Federal Reserve G.19	Average credit card rate	STATED
NCUA PALs II rule	PAL caps	STATED
EIA RECS 2020 microdata (CSV, ~18,500 households, weighted to 123.5M homes)	Opening finding	Codebook column names to pull
Best Buy Products API	New prices (cached)	Key requested; may not arrive in time
User-supplied listings, tags, leases	Used prices, rent-to-own terms	No scraping
EIA monthly Georgia residential prices	Rate projection (TigerData only)	Conditional

Opening finding (RECS 2020)

- Headline: energy burden (annual energy spending as a share of household income) by income bracket and renter status (KOWNRENT). Product-agnostic.

- Optional supporting line: share of households running equipment 20+ years old, across all equipment RECS covers, not one item.
- Use household weights; compute standard errors with the replicate weights; report sample size per cell. Label modeled consumption "EIA-estimated".
- Use Georgia if cell sizes hold up; otherwise the South region. Say which on the slide.

5. Build order (priority)

#	Deliverable	Done when
1	Cost engine + Pydantic models + pytest	All five paths, three headline numbers, source labels, 36-month flows; tests pass
2	One sourced category profile + manual entry	Typed model/serial produces a full receipt
3	Lease path	Scanned or typed lease shows total vs cash price and cheapest buyout week
4	Receipt UI with tap-to-source	Every number opens source + formula
5	Grok vision scan (HTTPS, real phone)	Scan equals manual entry for the demo cards
6	RECS finding	Weighted table + chart with sample sizes
7	Used vs new ranking + Grok request parsing (Visa)	Listing screenshot ranked against cached new offers by cost per year
8	Grok Voice read-aloud, English + Spanish	Receipt read aloud in both languages
9	Second category profile	Only if 1-8 are solid

6. Pitch and demo

Pitch outline (about 3 minutes)

- 1. The finding: lower-income households spend a larger share of income on energy, and "cheap" options make it worse.
- 2. Why: the price today hides running costs, interest, lease terms, and how long it lasts.
- 3. Demo: scan a label, a lease, a listing. Camera stays on the receipt, not the item.
- 4. Not a wrapper: type a model number by hand, get the identical receipt.
- 5. Trust: tap any number for its source and formula; rated vs estimated vs not estimated.
- 6. Close: the cost line and the carbon line together.

Tagline: **Every store sorts by sticker price. We sort by what it actually costs you.**

Expo table setup

- Printed cards of real rating labels, a printed rent-to-own listing, a used-listing screenshot. Phone mirrored to laptop. Upload button with saved images as fallback.
- Say plainly that the cards are printouts of real labels.
- Don't narrate what the item is; say "a rating label from a 2004 unit" and let the receipt talk. Two scans plus one typed entry is enough.
- Aim for about 5 seconds of visible AI to 85 of engine, but make the Grok shopping request visible once for Visa.

Stage accuracy rules (non-negotiable)

- "Effective annual cost" for rent-to-own. Never "APR".
- Never say a user qualifies for any assistance or loan.
- No absolute claims like "the cheapest option is always the most expensive". Say "can be", or only claim what the receipt on screen shows.
- The 10.2% Atlanta low-income energy burden figure (ACEEE) is 2016 data; say so, or use a newer figure if one exists.
- Cite energy figures to ENERGY STAR / DOE directly. Rent-to-own totals from the actual lease.
- Drop "HL Hunt Research Desk" (4.63M rent-to-own households): unverified.
- Avoid the Georgia PSC data center topic; it pulls the pitch toward policy.

7. VERIFY before coding as fact

These are working assumptions from planning, not confirmed facts.

- The DOE standard in force is a ceiling on rated energy when new; aged units may draw more. Refrigerator test procedures may have changed around 2014, so old and new ratings may not be directly comparable.
- Georgia Power marginal rate: tier, season and riders that apply per kWh.
- eGRID rate type: non-baseload may be more defensible than total output for avoided use.
- Water heating dominates some appliances' energy; gas vs electric changes results. Either one question or show both lines.
- Weather-driven items (cooling) need a documented usage-hours assumption; label estimated.
- RECS 2020 variable names, income variable, replicate weights, and Georgia cell sizes.
- Whether any Georgia regulator publishes rent-to-own multiples.
- PAL eligibility depends on credit union membership.
- Whether a newer ACEEE Atlanta energy burden figure exists.
- TigerData prize existence. Best Buy API key status.

8. Open items

- Confirm TigerData prize.
- Chase the Best Buy API key; otherwise hand-built cache of real listings with URLs and dates.
- Find real rating-label images for the demo and print them.
- Pull the Georgia Power residential tariff sheet and the RECS 2020 codebook.
- Start Notability usage and screenshots now.

9. Already rejected (do not re-suggest)

SONAR (NSA audio detector); hospital price transparency; prior-auth prediction; agentic-commerce price integrity; dark-pattern detection; dark-vessel AIS gaps; lead service line prediction; GreenGate prompt router. The international internship eligibility idea lives in a separate project.